Instructor's Note:

`bitwhiz` is the reference hydrocode for ME 697, Intermediate Shock and Detonation Theory (Purdue, Fall 2026): a one-dimensional, planar, Lagrangian code for plate impact, with shock and release waves crossing layers of solids and liquids, and spall where release waves meet in tension. Two solution methods share one code: this sheet is Method AV (artificial viscosity), and the companion sheet is Method RS (a Riemann solver). It was written by D. Kittell with the assistance of Claude Opus 5.5 (Anthropic), an AI model; the instructor directed, reviewed, and tested every part of it and is responsible for its content.

These equations build upon U3. Continuum Mechanics & Balance Laws and the Lecture 19 Addendum (artificial viscosity method)

Notation & Scope:

- One-dimensional hydrocode, planar, Lagrangian, and purely hydrodynamic, such that $\boldsymbol{\sigma_{xx} = -(P + q)}$ with $\boldsymbol{Y = 0}$ and artificial viscosity, $\boldsymbol{q}$
- Grid discretization has integers, $\boldsymbol{k}$, at cell *centers*, and $\boldsymbol{k \pm \frac{1}{2}}$ at *faces*; time step index is $\boldsymbol{n}$: whole and half steps considered, $\boldsymbol{\Delta t^n}$ spans $\boldsymbol{n \to n+1}$
- Staggered Grid, $\boldsymbol{u}$, $\boldsymbol{x}$ face-centered; $\boldsymbol{\rho}$, V $= \boldsymbol{1/\rho}$, $\boldsymbol{e}$, $\boldsymbol{P}$, $\boldsymbol{q}$, $\boldsymbol{c}$ cell-centered
- Cell Mass, $\boldsymbol{M_k}$ is mass per unit area of cell $\boldsymbol{k}$, fixed for all time (note: the areal mass coordinate $\boldsymbol{h}$ of U7 advances by $\boldsymbol{M_k}$ per cell)
- Sound Speed, $\boldsymbol{c_k}$ is the isentropic sound speed of cell $\boldsymbol{k}$ (subscript $\boldsymbol{s}$ dropped); initial state is $(\ )_{\boldsymbol{0}}$, Hugoniot state is $(\ )_{\boldsymbol{H}}$
- Boxes titled **bitwhiz:** go beyond the ***project rules***, to what the reference code `bitwhiz` does

A Consistent Set of Units (no conversion factors anywhere below):

- $\boldsymbol{x}$ [mm], $\boldsymbol{t}$ [$\boldsymbol{\mu}$s], $\boldsymbol{\rho}$ [g/cm$^{\boldsymbol{3}}$], $\boldsymbol{u}$, $\boldsymbol{c}$ [km/s] = [mm/$\boldsymbol{\mu}$s], $\boldsymbol{P}$, $\boldsymbol{q}$ [GPa], $\boldsymbol{e}$ [MJ/kg] = [km$^{\boldsymbol{2}}$/s$^{\boldsymbol{2}}$]
- Cell Mass, $\boldsymbol{M_k}$ [g/cm$^{\boldsymbol{3}}$·mm] e.g., $\dfrac{\boldsymbol{\Delta t\, \Delta P}}{\boldsymbol{M}} \to \dfrac{[\boldsymbol{\mu}\text{s}][\text{GPa}]}{[\text{g/cm}^{\boldsymbol{3}}][\text{mm}]} = [\text{km/s}]$
- If you stick to using these units, there won't be any conversion factors inside your code. If you change units, e.g., $\boldsymbol{x}$ in [cm] instead, then there will be conversion factors.

Step 0. Mesh and Initial Conditions:

- Faces, $\boldsymbol{x^0_{k+\frac{1}{2}}}$ uniform within each material; adjust zones per material so that ***every material interface lands on a cell face***
- Cell Mass, $\boldsymbol{M_k = \rho_{0,k}\left(x^0_{k+\frac{1}{2}} - x^0_{k-\frac{1}{2}}\right)}$ computed once, constant over time
- Cell State, $\boldsymbol{\rho^0_k = \rho_{0,k}}$, $\boldsymbol{e^0_k = 0}$, $\boldsymbol{P^0_k = 0}$, $\boldsymbol{q^0_k = 0}$, $\boldsymbol{c^0_k = c_{0,k}}$
- Face Velocity, $\boldsymbol{u^{-\frac{1}{2}}_{k+\frac{1}{2}}}$ the initial velocity of the material on both sides; on an impact face you may use the momentum-weighted value $\boldsymbol{\dfrac{M_k u_{0,k} + M_{k+1} u_{0,k+1}}{M_k + M_{k+1}}}$

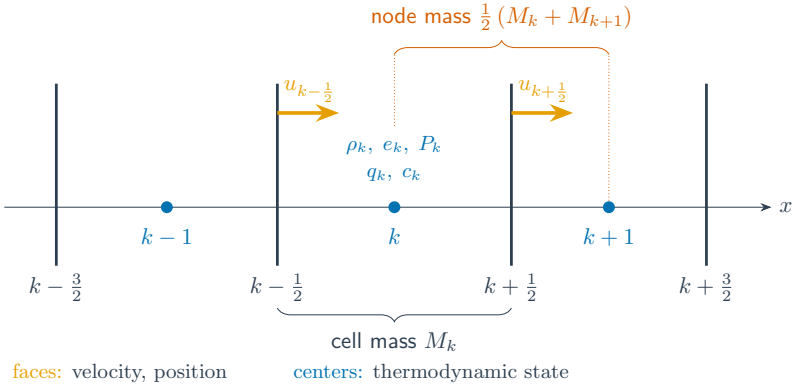


Figure 1: Staggered grid (cartoon). Velocity and position live on the faces; the thermodynamic state lives at the centers. The momentum update divides by the node mass, the cell-to-cell average shown in red.

Step 1. Velocity, Position, and Density:

- Momentum, for every face between cells $\boldsymbol{k}$ and $\boldsymbol{k+1}$

$$\boldsymbol{u_{k+\frac{1}{2}}^{n+\frac{1}{2}} = u_{k+\frac{1}{2}}^{n-\frac{1}{2}} + \frac{2\,\Delta t^n}{M_k + M_{k+1}}\left[\left(P_k^n - P_{k+1}^n\right) + \left(q_k^n - q_{k+1}^n\right)\right]}$$

- Cell Kinematics, $\boldsymbol{x_{k+\frac{1}{2}}^{n+1} = x_{k+\frac{1}{2}}^{n} + u_{k+\frac{1}{2}}^{n+\frac{1}{2}}\,\Delta t^n}$ the mesh moves with mass

- Cell Density, $\boldsymbol{\rho_k^{n+1} = \dfrac{M_k}{x_{k+\frac{1}{2}}^{n+1} - x_{k-\frac{1}{2}}^{n+1}}}$ and, if needed, $\mathrm{V}_k^{n+1} = 1/\rho_k^{n+1}$

- Note: the velocity step spans $\boldsymbol{n-\frac{1}{2} \to n+\frac{1}{2}}$. With a fixed $\boldsymbol{\Delta t}$ this is exactly $\boldsymbol{\Delta t^n}$; when $\boldsymbol{\Delta t}$ changes every cycle, $\boldsymbol{\frac{1}{2}\left(\Delta t^{n-1} + \Delta t^n\right)}$ is the time-centered choice (see Fig. 2)

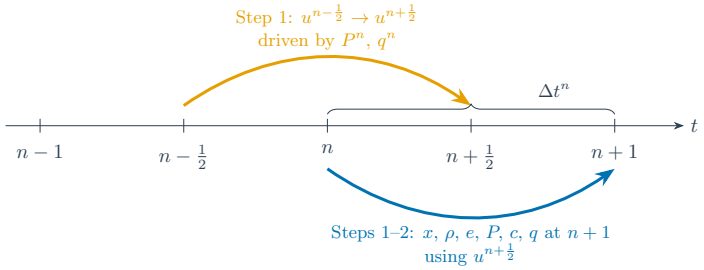


Figure 2: Leapfrog time integration (cartoon). Velocity is advanced on half steps by the state at the whole step between them; everything else is advanced on whole steps by the new velocity.

Step 2. Energy and Pressure:

- Total Energy (cell $\boldsymbol{k}$), face stresses are mass-weighted, face viscosities averaged

$$e_k^{n+1} = e_k^n + \frac{\Delta t^n}{M_k}\left[\frac{M_k P_{k-1}^n + M_{k-1} P_k^n}{M_k + M_{k-1}} + \frac{q_k^n + q_{k-1}^n}{2}\right] u_{k-\frac{1}{2}}^{n+\frac{1}{2}}$$

$$- \frac{\Delta t^n}{M_k}\left[\frac{M_{k+1} P_k^n + M_k P_{k+1}^n}{M_k + M_{k+1}} + \frac{q_k^n + q_{k+1}^n}{2}\right] u_{k+\frac{1}{2}}^{n+\frac{1}{2}}$$

$$+ \frac{1}{8}\left[\left(u_{k+\frac{1}{2}}^{n-\frac{1}{2}} + u_{k-\frac{1}{2}}^{n-\frac{1}{2}}\right)^2 - \left(u_{k+\frac{1}{2}}^{n+\frac{1}{2}} + u_{k-\frac{1}{2}}^{n+\frac{1}{2}}\right)^2\right]$$

- Reading the terms: work done by the face stresses, minus the change in the cell's kinetic energy $\frac{1}{2}\bar{u}^2$, with $\bar{u} = \frac{1}{2}\left(u_{k-\frac{1}{2}} + u_{k+\frac{1}{2}}\right)$ ***easiest place for mistakes***

- Pressure, $P_k^{n+1} = P\left(\rho_k^{n+1}, e_k^{n+1}\right)$ from the equation of state of cell $\boldsymbol{k}$

- Pressure Bounds, $P \geq 0$ flag the first cell that goes tensile and the time it happens, then set it to zero and continue (project Sec. 3; `bitwhiz`: see Tension and Cell Breaking)

bitwhiz: Tension and Cell Breaking (Spall):

- Spall Strength, $\boldsymbol{P_{sp} > 0}$ per material (deck: `spall_strength`, default 0.01 GPa); Damage, $\boldsymbol{\mathcal{D}_k = 0}$ intact, $\boldsymbol{\mathcal{D}_k = 1}$ broken; every cell starts intact
- Intact, keep the EOS pressure while $\boldsymbol{P_k^{n+1} \geq -P_{sp}}$ an intact cell ***holds tension***
- Breaking, if intact and $\boldsymbol{P_k^{n+1} < -P_{sp}}$: $\boldsymbol{\mathcal{D}_k = 1}$ and $\boldsymbol{P_k^{n+1} = 0}$ it spalls on this step
- Broken, $\boldsymbol{P_k^{n+1} = \max\left(P_k^{n+1},\, 0\right)}$ on every later step it can be compressed and shocked, never pulled
- Project Rule, $\boldsymbol{P_{sp} = 0}$ gives the $\boldsymbol{P \geq 0}$ of Step 2, up to a round-off floor: a cell breaks only below $\boldsymbol{-10^{-9}\rho_0 c_0^2}$
- Void Cell, a broken cell stretched until $\boldsymbol{c^2 \leq 0}$ at $\boldsymbol{P = 0}$ (not reached by the class materials): $\boldsymbol{P = q = c = 0}$, $\boldsymbol{\Delta t_k = \dfrac{\Delta x}{|\bar{u}|}}$; it keeps its mass and momentum, and the EOS resumes if it is compressed
- Order, EOS pressure, then this rule, then $\boldsymbol{c_k^{n+1}}$ ***at the pressure kept***, then $\boldsymbol{q_k^{n+1}}$

Equations of State, $\boldsymbol{P(\rho, e)}$:

- Mie-Grüneisen, $\boldsymbol{P(\mathrm{V}, e) = P_H(\mathrm{V}) + \rho_0\Gamma_0\left[e - e_H(\mathrm{V})\right]}$ with $\boldsymbol{\rho\Gamma = \rho_0\Gamma_0}$
- Hugoniot Reference, $\boldsymbol{P_H = \dfrac{u_p^2}{\mathrm{V}_0 - \mathrm{V}} = \rho_0\,\eta\,U_S^2}$ $\boldsymbol{e_H = \dfrac{1}{2}u_p^2}$ for $\boldsymbol{P_0 = 0}$, $\boldsymbol{e_0 = 0}$ (the last form has no $\boldsymbol{0/0}$ at V_0 using the code's $\boldsymbol{\eta}$ definition, below)
- Compression, $\boldsymbol{\eta = 1 - \dfrac{\mathrm{V}}{\mathrm{V}_0} = \dfrac{\rho - \rho_0}{\rho}}$ $\boldsymbol{u_p = \eta\,U_S(\eta)}$ with $\boldsymbol{U_S = c_0 + s_1 u_p + (s_2/c_0)\,u_p^2}$ inverted as

  . $$\boldsymbol{U_S(\eta) = \frac{2c_0}{(1 - s_1\eta) + \sqrt{(1 - s_1\eta)^2 - 4s_2\eta^2}}} \quad \text{for } \boldsymbol{\eta \geq 0}$$

- Note: the $\boldsymbol{u_p}$ above is the *Hugoniot* particle velocity at the cell's current volume, ***not*** the cell's material velocity $\boldsymbol{u}$
- Note: code $\boldsymbol{\eta}$ as $\boldsymbol{(\rho - \rho_0)/\rho}$, exactly 0 at $\boldsymbol{\rho_0}$; $\boldsymbol{1 - \rho_0\,(1/\rho_0)}$ is not, for 15% of densities

bitwhiz: The Same EOS, Split, and Its Expansion Branch ($\mathbf{V > V_0}$):

- Split Form, $\boldsymbol{P(\mathrm{V}, e) = \rho_0 \Gamma_0 e + \Pi(\mathrm{V})}$ possible because $\boldsymbol{(\partial P/\partial e)_{\mathrm{V}} = \rho_0 \Gamma_0}$ is a constant $(\boldsymbol{\rho\Gamma = \rho_0 \Gamma_0})$

- Compression, $\boldsymbol{\Pi = P_H - \rho_0 \Gamma_0 e_H}$ $\boldsymbol{\dfrac{d\Pi}{d\mathrm{V}} = \dfrac{dP_H}{d\mathrm{V}} - \rho_0 \Gamma_0 \dfrac{de_H}{d\mathrm{V}}}$ the Mie-Grüneisen form above, rearranged

- Expansion, $\boldsymbol{U_S}$–$\boldsymbol{u_p}$ data exist only in compression, so for $\boldsymbol{\eta < 0}$ `bitwhiz` uses a cubic in the density change $\boldsymbol{\chi = \rho - \rho_0 < 0}$

  . $\boldsymbol{\Pi = a_1 \chi + \dfrac{a_2}{2}\chi^2 + \dfrac{a_3}{6}\chi^3}$ $\boldsymbol{\dfrac{d\Pi}{d\mathrm{V}} = -\rho^2 \left(a_1 + a_2 \chi + \dfrac{a_3}{2}\chi^2 \right)}$

- Coefficients, chosen so that $\boldsymbol{\Pi}$ and its first three derivatives match the compression branch at V_0

  . $\boldsymbol{a_1 = c_0^2}$ $\boldsymbol{a_2 = \dfrac{c_0^2}{\rho_0}(4s_1 - 2 - \Gamma_0)}$

  . $\boldsymbol{a_3 = \dfrac{6c_0^2}{\rho_0^2}\left(3s_1^2 - 4s_1 - \Gamma_0 s_1 + \Gamma_0 + 2s_2 + 1\right)}$

- Result, $\boldsymbol{P}$, $\boldsymbol{c}$, and the first two derivatives of $\boldsymbol{c}$ are continuous through V_0; a polynomial, so it can be evaluated at ***any*** expansion (a `bitwhiz` rule, not from the literature)

- Note: how much tension a material holds is ***not*** the EOS's business; that is $\boldsymbol{P_{sp}}$ (Tension and Cell Breaking)

Isentropic Sound Speed, c_k^{n+1}:

- Definition, $c^2 = -\mathrm{V}^2 \left(\frac{\partial P}{\partial \mathrm{V}}\right)_s$ evaluated cell by cell, at a point, after the pressure update
- Mie-Grüneisen, $c_k^{n+1} = \frac{1}{\rho_k^{n+1}} \sqrt{\rho_0 \Gamma_0 \left[P_k^{n+1} + \frac{de_H}{d\mathrm{V}}\right] - \frac{dP_H}{d\mathrm{V}}}$
- Helpful Identity, $\frac{de_H}{d\mathrm{V}} = \frac{1}{2}\left[(\mathrm{V}_0 - \mathrm{V})\frac{dP_H}{d\mathrm{V}} - P_H\right]$ both at V_k^{n+1}, so only $dP_H/d\mathrm{V}$ is needed
- Check, at $\mathrm{V} = \mathrm{V}_0$ and $e = 0$ the formula returns $c = c_0$; the derivative of $U_S(\eta)$ is the hardest algebra in the whole code
- Hugoniot Slope, $\frac{dP_H}{d\mathrm{V}} = -\rho_0^2 \left(U_S^2 + 2\eta U_S \frac{dU_S}{d\eta}\right)$

  . $\frac{dU_S}{d\eta} = \frac{2c_0\left[s_1 + (s_1 - s_1^2\eta + 4s_2\eta)/\xi\right]}{(1 - s_1\eta + \xi)^2}$ $\xi = \sqrt{(1 - s_1\eta)^2 - 4s_2\eta^2}$
- Any Branch (`bitwhiz`), $c^2 = \mathrm{V}^2\left[\rho_0\Gamma_0 P - \frac{d\Pi}{d\mathrm{V}}\right]$ the same formula, rearranged; P is the pressure kept by the spall rule
- Trap: $dP_H/d\mathrm{V}$ (along the Hugoniot) is *not* $(\partial P/\partial \mathrm{V})_e$; mixing them gives c 4–9% high at $\eta = 0.1$ (class materials), yet both give $c = c_0$ at V_0

Artificial Viscosity, q_k^{n+1}:

- Velocity Jump, $\Delta u_k = u_{k-\frac{1}{2}}^{n+\frac{1}{2}} - u_{k+\frac{1}{2}}^{n+\frac{1}{2}}$ positive when the cell is being compressed
- Logical check, $q_k^{n+1} = 0$ unless $\Delta u_k > 0$ and $\rho_k^{n+1} - \rho_k^n > 0$
- Linear and quadratic, $q_k^{n+1} = B_L \rho_k^{n+1} c_k^{n+1} |\Delta u_k| + B_Q \rho_k^{n+1} (\Delta u_k)^2$
- Defaults, $B_L = 0.1$ and $B_Q = 2$ free to vary; too little rings behind the shock, too much smears it

Time Step (CFL Condition):

- Per Cell, $\boldsymbol{\Delta t_k} = \dfrac{x_{k+\frac{1}{2}}^{n+1} - x_{k-\frac{1}{2}}^{n+1}}{\left|\frac{1}{2}\left(u_{k-\frac{1}{2}}^{n+\frac{1}{2}} + u_{k+\frac{1}{2}}^{n+\frac{1}{2}}\right)\right| + c_k^{n+1}}$ looks like $\dfrac{\Delta x}{|u| + c}$
- CFL Criterion, $\boldsymbol{\Delta t_{CFL}} = \min_k (\Delta t_k)$ the smallest cell sets the pace
- In the code, $\boldsymbol{\Delta t^{n+1}} = \mathbf{0.6}\ \boldsymbol{\Delta t_{CFL}}$ 0.6 is a safety factor (`bitwhiz`: deck key `CFL`, default 0.6)
- First Step, $\boldsymbol{\Delta t^0}$ from the $\boldsymbol{t = 0}$ state ($\boldsymbol{c = c_0}$), with $\boldsymbol{\Delta t^{-1} = \Delta t^0}$ Last Step, shortened to land exactly on the termination time

Boundary Conditions (Void on Both Ends):

- Ghost Cells, $\boldsymbol{k = 0}$ and $\boldsymbol{k = k_{max} + 1}$ with $\boldsymbol{M = P = q = 0}$ (see Fig. 3)
- Effect, the same Step 1 and Step 2 formulas apply at faces $\frac{1}{2}$ and $\boldsymbol{k_{max} + \frac{1}{2}}$: the node mass is half a real cell and the face stress is zero, which is a ***free surface*** surrounded by an infinite zero-pressure void
- Note: in the energy update the end face's stress is $\frac{1}{2}\boldsymbol{q_k}$, not zero, while the end cell is being compressed ($\boldsymbol{q_k > 0}$); that is what the Step 2 formula gives with a ghost, and what `bitwhiz` does
- Interfaces, nothing special: two materials meet at a shared face, and each cell calls its own EOS

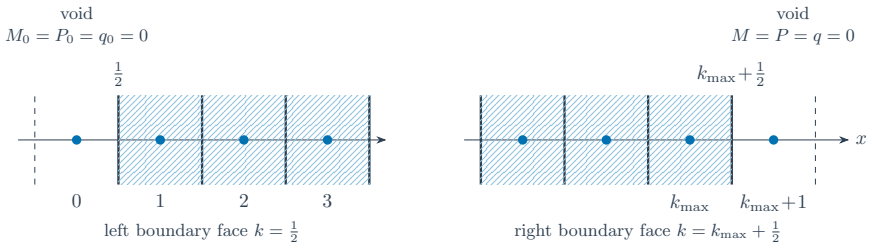


Figure 3: Void boundaries (cartoon). One empty ghost cell on each end of the mesh; the real material is hatched.

One Cycle, In Order:

a. Velocity $\boldsymbol{u^{n+\frac{1}{2}}}$ on every face, from $\boldsymbol{P^n}$, $\boldsymbol{q^n}$

b. Positions $\boldsymbol{x^{n+1}}$, then densities $\boldsymbol{\rho^{n+1}}$

c. Energies $\boldsymbol{e^{n+1}}$, from $\boldsymbol{P^n}$, $\boldsymbol{q^n}$, $\boldsymbol{u^{n\pm\frac{1}{2}}}$

d. Pressures $\boldsymbol{P^{n+1}}$ (keep them $\geq \mathbf{0}$; `bitwhiz`: the spall rule), then sound speeds $\boldsymbol{c^{n+1}}$

e. Viscosities $\boldsymbol{q^{n+1}}$, then the next time step $\boldsymbol{\Delta t^{n+1}}$

f. Write output if needed; $\boldsymbol{n \leftarrow n+1}$; repeat until the termination time

- Check, total mass $\sum_k \boldsymbol{M_k}$ is exact; total energy $\sum_k \boldsymbol{M_k}\left(\boldsymbol{e_k} + \frac{1}{2}\boldsymbol{\bar{u}_k^2}\right)$ and momentum $\sum_k \boldsymbol{M_k \bar{u}_k}$ should hold to round-off and truncation error (`bitwhiz` reports both every cycle)

bitwhiz: Where This Lives in the Code:

- Steps, Step 0 → `maincode` sections 5–6; Step 1 → `face_step` (loop 1) and the density in `cell_step` (loop 2); Step 2 energy → `cell_step`; EOS, spall rule, $\boldsymbol{c}$, $\boldsymbol{q}$, $\boldsymbol{\Delta t_k}$ → `cell_state`, `cell_dt` (loop 3); EOS formulas → `eos_mod`; the cycle and $\boldsymbol{\Delta t_{CFL}}$ → `maincode` section 7

- Database Slots, one array `database(nvar,imax)`; faces and cells alternate along the second index

On the faces		In the cells		Material (cells)	
$\boldsymbol{x^{n+1}}$	`jx`	$\boldsymbol{M_k}$	`jm`	$\boldsymbol{\rho_0}$	`jr0`
$\boldsymbol{x^n}$	`jx_old`	$\boldsymbol{\rho^{n+1}}$, $\boldsymbol{\rho^n}$	`jrho`, `jrho_old`	$\boldsymbol{c_0}$	`jc0`
$\boldsymbol{u^{n+\frac{1}{2}}}$	`ju`	$\boldsymbol{e^{n+1}}$, $\boldsymbol{e^n}$	`je`, `je_old`	$\boldsymbol{s_1}$, $\boldsymbol{s_2}$	`js1`, `js2`
$\boldsymbol{u^{n-\frac{1}{2}}}$	`ju_old`	$\boldsymbol{P}$, $\boldsymbol{q}$, $\boldsymbol{c}$	`jp`, `jq`, `jc`	$\boldsymbol{\Gamma_0}$	`jg0`
node mass	`jmf`	$\boldsymbol{\mathcal{D}}$	`jdam`	$\boldsymbol{P_{sp}}$	`jsp`

- Index Rule, cell $\boldsymbol{k}$ is at $\boldsymbol{i = 2k+1}$, face $\boldsymbol{k+\frac{1}{2}}$ at $\boldsymbol{i = 2k+2}$; ghost cells are $\boldsymbol{k = 0}$ and $\boldsymbol{k_{\max}+1}$ (`shock_mod`: `icell`, `iface`)

- Units, exactly the set above; `bitwhiz` has no conversion factors